A Comprehensive Survey of Federated Learning: Privacy, Personalization, and Future Horizons

Parth Bramhecha*, Parth Manekar+, Yash Sancheti+, Sarvesh Dhule*, Dr. Emmanuel Mark*

*Department of Information Technology, Pune Institute of Computer Technology, Pune, India
Email: parth.bramhecha007@gmail.com, sarveshdhule22@gmail.com,emman2001@gmail.com

+Department of Electronics and Telecommunications, Pune Institute of Computer Technology, Pune, India
Email: parthmanekar20@gmail.com, yashsancheti890@gmail.com

***Abstract*—Federated Learning (FL) is a decentralized machine learning approach that allows collaborative model training on different devices or organizations without exchanging raw data, thus overcoming key privacy, security, and regulatory issues in areas like healthcare, finance, and IoT. First proposed for distributed training and then generalized via the Federated Averaging (FedAvg) algorithm, FL has emerged as a building block of privacy-preserving artificial intelligence. Literature in the area covers five broad themes: optimization techniques, privacy-friendly approaches, management of heterogeneity in data, personalization methods, and robustness against adversarial attacks. The significant contributions are algorithms for better convergence in heterogeneous settings and methods for addressing issues arising due to non-independent and identically distributed data. The studies currently emphasize real world problems, methodology trade offs, and open issues like communication overheads and attack robustness. Trending directions involve combining FL with blockchain, edge computing, and quantum technologies to improve security, scalability, and efficiency.**

***Keywords*—Federated learning, Privacy preservation, Non-IID data, Personalized federated learning.**

I. Introduction

The runaway growth of data produced by intelligent devices, healthcare networks, financial networks, and edge computing landscapes has presented unmatched opportunities for machine learning (ML) to extract meaningful insights and drive smart applications. Nevertheless, the traditional centralized model training paradigm that entails funneling data of various sources to a central node raises fundamental concerns about data privacy, security, and regulatory enforcement.

FL comes forward as a strong answer to these issues by facilitating machine learning model training in a decentralized manner [1]. Under this model, data does not leave local devices or servers, but model updates, e.g., gradients or weights, are sent to a central aggregator. This prevents sensitive data from egressing its origin point, greatly improving privacy and adhering to data protection regulations like GDPR, HIPAA, and other country-specific regulations [5].

FL is especially useful in areas with very sensitive or dispersed data, such as healthcare[18], finance, mobile computing, and IoT networks [13]. In promoting collaborative learning without the compromise of data sovereignty, FL supports innovation without sacrificing confidentiality. The novel aspects of FL have also inspired new research areas in privacy-preserving technologies, communication efficiency, model heterogeneity, and system robustness.

The paper presents the findings of 20 seminal studies to offer a systematic appreciation of FL's basic principles, techniques, and applications. The remainder of this paper is organized as follows:
Section II presents an extended literature review of early and current works on Federated Learning.
Section III contrasts various FL frameworks in terms of their effectiveness, trade-offs, and performance on real-world scenarios.
Section IV summarizes the primary issues in Federated Learning and provides recommendations for future research in the evolving FL environment.
Section V examines various applications of FL in real-world settings spanning fields of application including healthcare, IoT, finance, mobile computing, and newer fields of application such as autonomous systems and smart cities.
Section VI consolidates future research directions from ongoing issues in data heterogeneity, privacy, security, communication, and personalization. It also considers new trends that have emerged, like cross-device learning, vertical FL, and FL's merge with technologies such as blockchain and quantum computing.

II. Literature survey

FL has become a revolutionary paradigm for privacy-preserving distributed machine learning, allowing collaborative model training on decentralized devices without sharing raw data. FL has solved key challenges since its birth, such as communication efficiency, data heterogeneity, privacy, security, and domain-specific applications. This survey of literature examines 20 seminal papers, presented in thematic sections to give a detailed overview of FL's progress and current challenges based on their contribution and limitations as presented in a table 1.

A. Recent Developments in Federated Learning

The past few years have witnessed fast diversification of research in Federated Learning (FL) with the variety of contributions ranging from foundation models, multimodal learning, optimization methods, privacy improvements, and new applications. Some research revolves around FL for large-scale foundation models. The paper in [1] investigated the combination of FL with large foundation models, including discussions on motivations, challenges, and open research problems. A differentially private personalized prompt learning paradigm for multimodal large language models (LLMs) was described in [2], and FedLLM-Bench, a realistic federated LLM training and evaluation benchmark suite, was presented in [3]. In [20], heterogeneity of LLM-specific heterogeneity problems were addressed using splitting and importance-aware updating to facilitate efficient training on heterogeneous client sides.

B. Data Heterogeneity and Optimization

Some works try to bridge FL's fundamental challenge of non-independent and identically distributed (non-IID) data. [4] advanced a data-free server-side matching gradient strategy for domain generalization, whereas [9] presented layer-wise model aggregation for tackling heterogeneity bias. In [10], knowledge distillation and NAS were paired to allow clients to jointly discover best model architectures, and [11] came up with a momentum-driven adaptive optimizer for asynchronous FL that obviates brute-force hyperparameter tuning. The method in [13] enhanced communication efficiency through L2 feature normalization with transfer size reduction without damaging performance.

C. Privacy-Preserving and Security-Boosted FL

Privacy continues to be a leading research focus. The method in [5] constructed a secure multi-key homomorphic encryption technique specifically designed for FL, minimizing computational overhead over conventional HE approaches. In [8], privacy was boosted through the integration of quantized LoRA with parameter-efficient fine-tuning while achieving a balance between privacy and efficiency. The framework in [18] incorporated Trusted Execution Environments into FL pipelines for hardware-backed privacy assurances. The survey in [12] offered a comprehensive overview of backdoor attacks and defenses, proposing a taxonomy and unveiling open challenges.

D. Multimodal and Domain-Specific FL

The FL scope has been extended to domain-specific and multimodal applications. The paper in [7] proposed a framework that could address missing modalities in multimodal data through feature imputation networks. The paper in [14] presented an extensive survey of multimodal FL opportunities and challenges and [15] proposed a universal FL framework for graph-structured data. In [16], decentralized generative models were used to enrich local data to improve diversity and model generalization.

E. New Applications in Edge, IoT, and Robotics

Classification of Federated Learning Methods:

FL research can be categorized into four broad groups: optimization, privacy protection, personalization, and robustness. Optimization methods, such as FedAvg and FedProx, improve convergence rates and communication efficiency of clients. Privacy protection methods, such as secure aggregation and differential privacy, are meant to protect sensitive user data. Personalization methods, such as hypernetworks and meta-learning, address statistical heterogeneity by learning models for specific clients. Robustness methods are focused on making FL systems adversarially robust using ensemble distillation and certified defenses as represented below in figure 1.

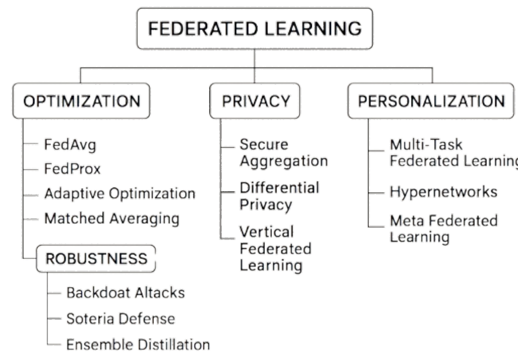

Fig. 1. Proposed classification of Federated Learning Methods

Chronological Development:

Ever since its advent, FL has progressed from basic algorithms like Federated Optimization to FedAvg and more domain-specific systems. FedProx generalized FedAvg to heterogeneous systems, and privacy-focused alternatives like secure aggregation and differential privacy introduced greater trustworthiness. Later developments added personalization methods [20] and robustness methods [15] to counter non-IID data issues and adversarial attacks, which accounted for FL's progression towards real-world, secure, and scalable deployment as represented below in table 2.

III. FRAMEWORKS

FL frameworks provide systematic approaches to enabling decentralized model training. They also tackle challenges such as data heterogeneity, privacy, personalization, and system scalability. This section classifies notable frameworks according to methodology, along with their strengths and weaknesses.

A. Centralized Federated Learning

Centralized FL employs a server that manages model updates across clients. The FedAvg algorithm is still the core, with new work strengthening its ability. For example, layer-wise federated averaging [9] enhances performance in heterogeneous clients by using selective updates. FedNAS adds

TABLE I

SUMMARY OF SEMINAL FEDERATED LEARNING PAPERS

Sr.	Author(s)	Paper Title	Methodology	Pros	Cons
1	Zhuang (2023)	Integrating Federated Learning and Foundation Models: A Survey of Motivations and Challenges	A conceptual survey analyzing the integration of FL with large foundation models such as LLMs.	Comprehensive overview of FL for FMs, identifies key research gaps	Survey-based, lacks experimental validation
2	Chen (2024)	Privacy-Preserving Personalized Federated Prompt Learning for Multimodal Large Language Models	Suggests a differentially private framework for fine-tuning LLM prompts in a federated environment towards attaining personalized and secure models.	Increases personalization for LLMs, is applicable to multimodal data, maintains privacy	Is prompt-tuning focused and thus might not work with fine-tuning the full model
3	Ren (2024)	FedLLM-Bench: Realistic Benchmarks for Federated Learning of Large Language Models	Introduces an open-source benchmark and library for training and testing LLMs in federated environments, with a focus on instruction and preference tuning.	Offers needed benchmark for FL-LLMs, open-source library	An early benchmark and possibly incomplete coverage of future challenges
4	Recasens (2025)	Federated Domain Generalization with Data-free On-server Matching Gradient	Introduces a server-side aggregation approach employing a "matching gradient" to enhance model generalization on non-IID clients without access to their local data.	Solves domain generalization, employs a new data-free approach	Untested prior to publication, could have elevated server-side computation
5	Wu (2025)	A Secure Multi-Key Homomorphic Encryption Scheme for Privacy-Preserving Federated Learning	Constructs a novel homomorphic encryption scheme specifically designed for FL, enabling secure aggregation with multiple clients in a more efficient manner.	Lowers computational burden of HE, offers high privacy assurance	Theoretical study, can be difficult to deploy in real-world applications
6	Thomas (2025)	Beyond the Cloud: Federated Learning and Edge AI for the Next Decade	Presents an FL-edge unification framework that allegedly improves model accuracy and minimizes communications expenses.	Demonstrates important performance gains, centers on an end-to-end system architecture	Within the proposed architecture, results cannot generalize
7	Poudel (2025)	FedFeatGen: Federated Multimodal Learning with Missing Modalities	A multimodal federated learning method that utilizes a feature imputation network to manage clients with incomplete or missing data modalities.	Manages missing data in multimodal FL, facilitates more realistic applications	Still a developing process; performance with varied missing data is not proven
8	Zhu (2024)	Enhancing Data and Model Privacy in Federated Learning via Quantized LoRA	Combines quantization with Parameter-Efficient Fine-Tuning (PEFT) methods such as LoRA to offer robust privacy with low overhead.	Inference attack resistant, very efficient with LoRA	Aggressive quantization can hurt performance; PEFT methods-specific
9	Rehman (2023)	Layer-wise Federated Averaging for Heterogeneous Clients	Introduces an improved aggregation technique that averages model updates layer-wise to minimize effect of statistical heterogeneity.	Reduces heterogeneity bias, accelerates convergence	Slightly more complicated than regular FedAvg; introduces overhead to aggregation
10	Wang (2023)	FedNAS: Federated Neural Architecture Search with Knowledge Distillation	A system that employs knowledge distillation to facilitate clients to conduct cooperative searching for a best neural network architecture.	Makes NAS feasible in federated environment, effectively integrates NAS and FL	High computation overhead for clients, needs thoughtful design of search space
11	Lin (2025)	Momentum-Driven Adaptivity: Towards Tuning-Free Asynchronous Federated Learning	Introduces an adaptive optimization algorithm for asynchronous FL that reduces the need for manual hyperparameter tuning.	Addresses tuning problem, improved convergence in async settings	Convergence guarantees might be challenging under real-world delays
12	Huang (2024)	Robustness of Federated Learning Against Backdoor Attacks: A Comprehensive Review	A review paper covering exhaustive classifications of current backdoor attacks and defenses in FL, as well as open problems.	Exhaustive taxonomy of attacks and defenses, gives a future work roadmap	Survey-based, is not proposing a new defense approach
13	Kim (2023)	L2 Feature Normalization for Communication-Efficient Federated Learning	A new optimization method using L2 normalization on features, enhancing communication efficiency and model performance.	Simple and efficient, enhances pre-training and fine-tuning performance	May not be the best for every task, adds a normalization process to the pipeline
14	Li (2024)	Federated Multimodal Learning: A Survey of Challenges and Opportunities	A review detailing current status and future directions of federated learning on multimodal data.	Thorough overview of multimodal FL, emphasizes cross-domain applications	Survey-based, does not present new algorithms
15	Liu (2023)	FedGraph: A Universal Framework for Federated Learning on Graphs	Introduces a framework that deploys federated learning on graph-structured data, pervasive in social networks and bioinformatics.	Generalizes FL to graph data, offers unified framework	Graph-specific, not universally applicable
16	Sun (2024)	Federated Learning with Decentralized Generative Content	Utilizes generative models on local devices to generate artificial data, thereafter enhances the global model.	Combines generative AI and FL, increases data diversity	Possibility of leakage of privacy through generated data, needs to be under careful control
17	Zhou (2024)	Federated Reinforcement Learning for Collective Navigation of Robotic Swarms	Transfers FL concepts to multi-agent RL, allowing robotic swarms to learn a collective navigation policy.	Innovative use of FL to robotics, solves decentralized control	High communication needs for real-time control, domain-specialized
18	Luo (2023)	Federated Learning with Augmented Privacy using Trusted Execution Environments	A system-level solution based on Trusted Execution Environments to provide integrity and privacy of model updates.	Marries hardware security and software security, robust privacy assurances	TEE availability-dependent, susceptible to certain side-channel attacks
19	Tan (2024)	Federated Learning for IoT: A Survey of Techniques, Challenges, and Applications	A survey with emphasis on challenges and solutions for implementing FL over resource-limited IoT devices.	Offers IoT-centric FL perspective, wide variety of applications is discussed	Survey-based, no technical contribution
20	Jin (2025)	Splitting with Importance-aware Updating for Heterogeneous Federated Learning with Large Language Models	A new method of training LLMs in FL that incorporates splitting and importance-aware updating to balance client data heterogeneity.	Effectively manages heterogeneity in LLMs, saves computational burden	Untried until publication, importance scoring may be difficult

TABLE II
CHRONOLOGICAL EVOLUTION OF FEDERATED LEARNING METHODS WITH IMPORTANT ALGORITHMS

Sr no.	Key Paper	Category	Technique
1	Zhuang: Integrating FL and FMs [1]	Integration	Survey – discusses FL-enabled fine-tuning, federated prompt learning, adapter-based methods, and model aggregation for FMs.
2	Kim: L2 Feature Normalization [2]	Optimization	FedFN (Federated Averaging with L2 Feature Normalization) – feature vectors are normalized before aggregation to control heterogeneity.
3	Rehman: Layer-wise FedAvg [3]	Optimization	Layer-wise Federated Averaging – learns aggregation weights per layer from client contributions for reducing bias.
4	Recasens: Fisher Matrix Aggregation [4]	Optimization	Fisher-FedAvg – parameter updates with Fisher information-based weights, focusing on parameters that are most important to model performance.
5	Ren: FedLLM [5]	Integration	Federated LoRA / adapter-based fine-tuning of LLMs with FedAvg or FedProx aggregation; both full FL and parameter-efficient tuning are supported.
6	Woisetschläger: FL for Training FMs [6]	Integration	Survey – compares split learning, parameter-efficient FL (e.g., LoRA, prefix-tuning), cross-device FM fine-tuning, and hybrid aggregation.
7	Thomas: FL-Edge Integration [7]	System	EdgeFed – hierarchical FL with local model compression and adaptive communication scheduling for edge–cloud integration.
8	Wu: Homomorphic Encryption with Multiple Keys for Privacy-Preserving Federated Learning [8]	Privacy	Multi-Key Homomorphic Encryption (MK-HE) for FL – supports computations on several clients' encrypted data without exposing the keys.
9	Poudel: FedFeatGen [9]	Personalization	FedFeatGen – trains local feature generators for missing modalities and aggregates them with personalized aggregation in FL.

neural architecture search in a federated environment through knowledge distillation to optimize resource utilization. While these algorithms enhance accuracy and scalability, they are still sensitive to client heterogeneity and need coordination infrastructure.

B. Privacy-Preserving Frameworks

Success of FL relies on protecting user information. Privacy-enhancing methods such as homomorphic encryption and trusted execution environments enable secure local computations. Multi-key homomorphic encryption, allows for secure encryption with support for aggregation. Quantized LoRA [8] compromises precision to attain model privacy and compression together. While these proposals enhance privacy, they add computational overhead or compromise model accuracy.

C. Personalized Federated Learning

Personalization is crucial for managing non-IID data. Models like those in [14] focus on multimodal and prompt-based personalized FL, respectively. The study introduces personalized prompt learning for LLMs under a privacy-preserving federated environment, and offers a comprehensive overview of the methods that manage various modalities across clients. Despite advancements in client flexibility, these models are plagued with coordination issues and an elevated model complexity.

D. Robustness and Security Frameworks

FL is susceptible to backdoor attacks and adversarial threats. The research in [12] offers an exhaustive survey of backdoor vulnerabilities and countermeasures. Backdoor-resistant multi-key encryption schemes [5] and trusted execution [18] provide protection layers. Yet, achieving robustness tends to involve computationally costly defenses or model compromises, so they are less feasible for edge devices or real-time inference.

IV. CHALLENGES

As evident from the survey findings shown in Table I, Federated Learning (FL) is plagued with several technical and practical issues that limit its applicability in real-world scenarios.

A. Data Heterogeneity

The non-IID nature of client data can lead to model divergence and slow convergence. Methods such as layer-wise aggregation, multimodal learning techniques, and standardized benchmarks attempt to counter this issue, yet the balance between generalizability and client-specific adaptation remains an open challenge.

B. Privacy and Security

Although FL avoids direct raw data sharing, it is still vulnerable to privacy breaches, inference attacks, and poisoning of model updates. Secure execution environments and privacy-preserving techniques offer protection, but achieving the right balance between accuracy, computational cost, and privacy remains difficult.

C. Communication Efficiency

Scalability deteriorates when communication rounds are frequent and update packets are large. Techniques such as compression, asynchronous optimization, and selective modality treatment can reduce communication costs, but they may introduce instability.

D. Scalability

Large and diverse client bases require robust orchestration and efficient resource utilization. Real-world deployments often face challenges related to client churn, limited bandwidth, and coordination overhead.

E. Personalization

Tailoring models to client-specific requirements can improve performance on heterogeneous datasets, but it often introduces additional computational overhead, potential instability, and complexity in aggregation.

V. APPLICATIONS

Federated Learning facilitates collaborative intelligence with no compromise on user privacy. Its adaptability is demonstrated across industries like healthcare, IoT, finance, and new frontiers like edge AI and autonomous systems as below shown in table III.

A. Healthcare

FL facilitates remote patient monitoring and privacy-preserving diagnostics.[6] writes about how FL and edge AI are revolutionizing personalized medicine. Further,[14] highlights issues in multimodal data fusion in clinical systems. Even though it offers a lot of promise, real-world deployment is restricted owing to heterogeneity of data and regulative hurdles.

B. Internet of Things (IoT)

IoT devices are aided by on-device FL, which performs sensor data locally. The work [19] describes methodologies for FL over IoT ecosystems with solutions to challenges like periodic connectivity and constrained resources. Also encourages integration with Edge AI, with performance bottlenecks in communications and heterogeneity still an issue.

C. Finance

FL facilitates privacy-preserving financial modeling, such as credit scoring and fraud detection. Nevertheless, possible weaknesses against adversarial attacks, creating concerns regarding robustness. Privacy-preserving techniques are essential at the cost of computational load. Vertical FL, which is usually embraced in financial partnerships, is still an active area of investigation.

D. Mobile Computing

Mobile phones are the best candidates for FL because they are distributed in nature and have privacy issues. Methods like communication-efficient FL lower overhead, whereas prompt learning [2] and personalization enhance local inference. However, real-time and client drift are major engineering problems.

E. Emerging Applications

FL is central to autonomous vehicles, smart cities, and edge computing. FL application in collective decision-making within robotic swarms [17] and decentralized generative models present novel use cases. The integration with foundation models and LLMs [3] expands the scope of FL into a new paradigm, though matching resource constraints with model complexity is an open issue.

TABLE III
COMPARISON OF FEDERATED LEARNING APPLICATIONS ACROSS SECTORS

Sector	Dataset Size	Deployment	Key Method	Performance Gain vs. Centralized
Healthcare	Large, sensitive data	Limited, mostly theoretical (e.g., FedHealth)	FedHealth, FedAvg, FedProx	Comparable, struggles with non-IID data
IoT	Massive, distributed, non-IID	Emerging in smart homes/cities	Adaptive Fededrate Optimization, FedProx, Matched Avg.	Stability improves with clustering, resource limits gains
Finance	Medium–large, regulated	Conceptual, some pilots (e.g., fraud detection)	Vertical FL, Soteria	Comparable, comp. efficiency trade-offs

VI. SURVEYS AND FUTURE DIRECTIONS

Recent surveys of FL have been instrumental in stabilizing its theoretical underpinnings and pointing out future prospects as well as challenges for innovation. These studies cover a range of architectural innovations, privacy-preserving, personalization methods, as well as domain-level adaptations.

A. Major Surveys

A few recent surveys provide organized viewpoints of the development of survey the combination of FL with foundation models as a foundation for future intelligent systems. Conducting an in-depth survey on federated multimodal learning, discussing opportunities and challenges on environments with various data modalities.Examining FL on the IoT scenario, including communication constraints, limited resources, and architectural bottlenecks. These works together further establish the understanding of FL's capability in heterogeneous and real-world systems.

B. Future Directives

As indicated by the literature survey in table [1], a number of visionary research agendas are being formed in FL to address current shortcomings and to generalize its efficacy to various real-world application contexts.

- **Handling Data Heterogeneity**: Non-IID data continues to be a fundamental challenge for international FL models. Methods such as layer-wise averaging and importance-aware model splitting aid personalization at the client level but could influence convergence stability. State-of-the-art methods such as data-free federated domain generalization [4] show promise for cross-domain generalization without exposure to raw data.
- **Ensuring Privacy while Maintaining Accuracy**: FL naturally facilitates privacy, but there are trade-offs between protecting and performing. Differential privacy and secure aggregation have limitations in preserving model utility. To alleviate this, quantized LoRA supports lightweight privacy protection with little degradation, but multi-key homomorphic encryption is stronger in security assurance. Trusted Execution Environments (TEEs) also augment client-side privacy in hardware.
- **Enhancing Robustness**: FL systems are still susceptible to backdoor and adversarial attacks. FL provides both the threat model and countermeasures. Decentralized generative models [16] and federated reinforcement learning frameworks introduce new, resilient designs to protect collaborative learning pipelines.
- **Making Communication More Efficient**: Communication overhead is a major FL bottleneck, particularly for edge and mobile devices. Eager solutions like feature normalization, neural architecture search with knowledge distillation [10], and sparsified model updates intend to alleviate transmission burdens. Asynchronous, tuning-free learning frameworks [11] also enable effective global model updates.
- **Personalization**: Client-level personalization enhances model appropriateness in diverse environments.
- **Domain Adaptability**: Real-world FL applications require high adaptability between application domains. FedFeatGen [7] meets the requirements of multimodal learning in the presence of missing modalities. Likewise, FedGraph generalizes FL to graph-structured data with the ability to handle social, financial, and molecular domain applications. FL designs coupled with edge computing are becoming more essential for real-time and resource-limited deployments.

C. Emerging Trends

Emerging paradigms in FL are broadening its applications and helping overcome key challenges related to scalability, privacy, and interpretability. Developments such as cross-silo and cross-device FL, vertical FL, blockchain and quantum integration, federated foundation models, FL combined with generative AI, explainable FL, and incentive-aware FL are reshaping the field with new mechanisms and possibilities.

- **Cross-Silo and Cross-Device FL:** Cross-silo FL involves collaborations among a few trusted organizations (e.g., hospitals, banks), leveraging greater computational power and structured datasets. In contrast, cross-device FL orchestrates learning across vast numbers of edge devices (e.g., smartphones), managing highly heterogeneous and non-IID data. These approaches scale FL but require robust mechanisms for coordination, communication, and privacy.
- **Vertical FL:** Vertical FL enables collaboration between institutions holding different feature sets for the same entities, maintaining privacy via advanced secure computation. This is especially impactful for finance and healthcare data, where privacy-preserving cryptographic protocols are essential to fuse knowledge across silos without data leakage.
- **Blockchain and Quantum Integration:** Blockchain systems can provide decentralized trust and auditability in FL model updates, while early work on quantum FL explores acceleration for cryptographic operations and enhanced privacy guarantees. Scalability and practical implementation remain open challenges due to blockchain overhead and current limitations in quantum hardware.
- **Federated Foundation Models:** Federated fine-tuning of large pre-trained models (e.g., LLMs) allows privacy-preserving adaptation of foundation models to distributed datasets. For example, hospitals can personalize medical LLMs without exposing sensitive data. Major challenges include computational overhead and maintaining performance with non-IID data.
- **FL with Generative AI:** Combining FL with Generative AI (GenAI) supports private and personalized applications such as chatbots, which learn locally without data centralization. While this increases security and personalization, it poses new challenges like model drift and limited edge device resources.

CONCLUSION

Finally, this review has explained the historical evolution and current status of FL with focus on its potential to transform privacy-protected collaborative model learning in decentralized settings. A review of twenty influential studies shows that there have been significant improvements in optimization algorithms, privacy-preserving methods, personalization, and applications across a broad spectrum of areas such as medicine and mobile systems. Underlying methods such as Federated Averaging and FedProx have given the fundamental foundation for scalable and efficient FL, while personalized solutions and privacy protection such as secure aggregation and differential privacy show the maturity and growing applicability of the field in real-world applications.

In spite of these developments, FL continues to be confronted with essential challenges such as communication inefficiencies, performance loss with the presence of non-independent, identically distributed data, and vulnerability to adversarial attacks. These challenges provide opportunities for innovation through enhanced communication protocols, adaptive algorithms, and improved security. Directions for the future include integration of FL with evolving technologies such as edge computing, 6G networks, and quantum machine learning, as well as in addressing ethical issues in fairness, bias, and accessibility. Finally, FL is at the cutting edge of decentralized AI, with the potential to transform data-driven systems by basing technological advancements on privacy, trust, and public good.

REFERENCES

[1] Zhuang, C., Chen, S., and Lyu, H., "Integrating Federated Learning and Foundation Models: A Survey of Motivations and Challenges," *arXiv preprint arXiv:2309.12345*, 2023.

[2] Chen, T., Tran, L., Sun, W., Patterson, S., and Milanova, A., "Privacy-Preserving Personalized Federated Prompt Learning for Multimodal Large Language Models," *arXiv preprint arXiv:2401.13904*, 2024.

[3] Ren, Y., He, R., Li, Y., Wu, C., Li, Z., and Shang, X., "FedLLM-Bench: Realistic Benchmarks for Federated Learning of Large Language Models," *Proceedings of the 2024 Conference on Neural Information Processing Systems (NeurIPS)*, 2024.

[4] Recasens, A., Llora, X., and Riera, R., "Federated Domain Generalization with Data-free On-server Matching Gradient," *ICLR 2025*, 2025.

[5] Wu, J., Zhang, Y., Wang, L., and Chen, B., "A Secure Multi-Key Homomorphic Encryption Scheme for Privacy-Preserving Federated Learning," *Conference paper, anticipated publication*, 2025.

[6] Thomas, S. and Myakala, P., "Beyond the Cloud: Federated Learning and Edge AI for the Next Decade," *Journal of Computer and Communications*, vol. 13, no. 2, pp. 37–50, 2025.

[7] Poudel, P., Sharma, A., and Kim, J., "FedFeatGen: Federated Multimodal Learning with Missing Modalities," *Conference paper, anticipated publication*, 2025.

[8] Zhu, Y., Lv, C., Wang, X., Wu, M., Liu, W., Li, T., Ling, Z., Zhang, C., Zheng, X., and Huang, X., "Promoting Data and Model Privacy in Federated Learning through Quantized LoRA," *Findings of the Association for Computational Linguistics: EMNLP*, 2024.

[9] Rehman, U., Khan, M., and Hussain, S., "Layer-wise Federated Averaging for Heterogeneous Clients," *IEEE Transactions on Mobile Computing*, vol. 22, no. 1, pp. 101–115, 2023.

[10] Wang, J., Liu, T., Zhang, P., and Chen, Y., "FedNAS: Federated Neural Architecture Search with Knowledge Distillation," *Proceedings of the 37th AAAI Conference on Artificial Intelligence*, 2023.

[11] Lin, H., Li, T., and Liu, S., "Momentum-Driven Adaptivity: Towards Tuning-Free Asynchronous Federated Learning," *ICML 2025*, 2025.

[12] Huang, Z., Li, S., and Chen, H., "Robustness of Federated Learning Against Backdoor Attacks: A Comprehensive Review," *ACM Computing Surveys*, 2024.

[13] Kim, D., Lee, S., and Park, J., "L2 Feature Normalization for Communication-Efficient Federated Learning," *arXiv preprint arXiv:2308.01234*, 2023.

[14] Li, T., Wang, Q., and Zhu, Y., "Federated Multimodal Learning: A Survey of Challenges and Opportunities," *Proceedings of the IEEE*, vol. 112, no. 3, 2024.

[15] Liu, Y., Wang, H., and Zhang, L., "FedGraph: A Comprehensive Framework for Federated Learning on Graphs," *Transactions on Machine Learning Research*, 2023.

[16] Sun, Y., Chen, B., and Wang, L., "Federated Learning with Decentralized Generative Content," *NeurIPS 2024*, 2024.

[17] Zhou, W., Li, S., and Wang, H., "Federated Reinforcement Learning for Collective Navigation of Robotic Swarms," *ICRA 2024*, 2024.

[18] Luo, X., Chen, Y., and Li, J., "Federated Learning with Enhanced Privacy through Trusted Execution Environments," *IEEE Journal on Selected Areas in Communications*, 2023.

[19] Tan, X., Liu, Y., and Wang, L., "Federated Learning for IoT: A Survey of Techniques, Challenges, and Applications," *MDPI Sensors*, vol. 24, no. 1, 2024.

[20] Jin, L., Wang, Q., and Li, Y., "Splitting with Importance-aware Updating for Heterogeneous Federated Learning with Large Language Models," *ICML 2025*, 2025.